

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL A-LEVEL

FURTHER MATHEMATICS

(9665/FM03) Unit FP2 Pure Mathematics

Tuesday 14 January 2025

07:00 GMT

Time allowed: 2 hours 30 minutes

Materials

- For this paper you must have the OxfordAQA Booklet of Formulae and Statistical Tables (enclosed).
- You may use a graphical calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.

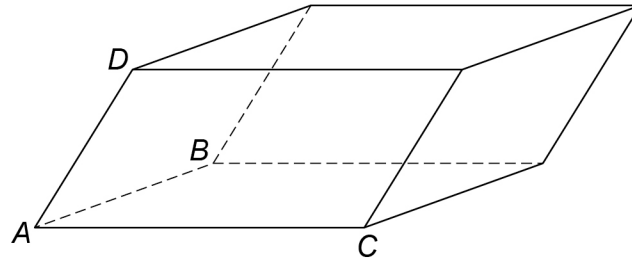
Advice

- Unless stated otherwise, you may quote formulae, without proof, from the booklet.
- Show all necessary working; otherwise marks may be lost.

For Examiner's Use	
Question	Mark
1	
2	
3	
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11	
12	
13	
14	
15	
TOTAL	



1 The diagram shows a parallelepiped and the four points A , B , C and D


$$\mathbf{a} = \begin{bmatrix} 3 \\ 4 \\ 2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} k+4 \\ 6 \\ 3 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 6 \\ 7 \\ 3 \end{bmatrix} \quad \text{and} \quad \mathbf{d} = \begin{bmatrix} k+5 \\ 3 \\ 5 \end{bmatrix}$$

It is given that the volume of the parallelepiped has a magnitude of 4 cubic units.

[6 marks]

[illegible]

6

Turn over ►



- 2 (a)** Use the definitions of $\cosh x$ and $\sinh x$ in terms of e^x and e^{-x} to show that

$$\cosh^2 x - \sinh^2 x = 1$$

[2 marks]

- 2 (b)** Use the result given in **part (a)** to solve the equation

$$\sinh^2 x - \cosh x - 5 = 0$$

Give any solutions as an exact natural logarithm.

[5 marks]



$\frac{\quad}{7}$

7

Turn over ►



3

$$AR = B$$

The matrix $\mathbf{R}$ represents a rotation about the x -axis through an angle θ

The matrix **B** is defined as

$$\mathbf{B} = \begin{bmatrix} 0 & \cos \theta & -\sin \theta \\ 1 & 0 & 0 \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

3 (a)

[3 marks]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

3 (b) Describe the single transformation represented by **A**

[2 marks]

5

Turn over for the next question

Turn over ►



- 4 (a) Explain why $\int_0^2 \frac{1}{\sqrt{4-x^2}} dx$ is an improper integral.

[1 mark]

- 4 (b) Evaluate $\int_0^2 \frac{1}{\sqrt{4-x^2}} dx$ showing the limiting process used.

[3 marks]

Answer _____



It is given that

Prove by induction that $f(n)$ is a multiple of 4 for all integers $n \geq 1$

[illegible]

5

6 (a) Show that

$$(r+1)^5 - (r-1)^5 = 10r^4 + 20r^2 + 2$$

[1 mark]

6 (b) Hence use the method of differences to show that

$$\sum_{r=1}^n r^4 = \frac{1}{30}n(n+1)(2n+1)(3n^2 + pn + q)$$

where p and q are integers

[7 marks]



[illegible]

8

7

$$r = \sec^2 \theta \sqrt{2(1 + \tan \theta)} \quad \text{where} \quad -\frac{\pi}{4} \leq \theta < \frac{\pi}{2}$$

The point A on C is where $\theta = 0$

The point B on C is where $\theta = \frac{\pi}{6}$

The point O is the pole.

Show that the area of the region bounded by the curve C and the lines OA and OB is

$$\frac{m + n\sqrt{3}}{108}$$

where m and n are integers.

[5 marks]

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[illegible]

5



$$\begin{array}{rrcr} 2x & - & y & + & 3z & = & 1 \\ x & + & (k-1)y & - & z & = & 3 \\ (k-3)x & - & y & + & z & = & 1 \end{array}$$

The planes **do not** meet at a unique point.

[3 marks]

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on the right side, suggesting it's resting on a surface.

Answer



8 (b) (i) Determine the number of solutions of the three equations when $k = 3$

Fully justify your answer.

[3 marks]

8 (b) (ii) Hence give a geometric interpretation for the three planes when $k = 3$

[1 mark]

7

Turn over ►



A curve is given parametrically by the equations

$$x = \sin \theta \cos \theta \quad \text{and} \quad y = \sin^2 \theta \quad \text{for} \quad 0 \leq \theta < \pi$$

The arc of the curve from $\theta = 0$ to $\theta = \frac{\pi}{6}$ is rotated through 2π radians about the x -axis to generate a surface with area S

Find the value of S

Give your answer in the form $\frac{\pi}{12}(p\pi + q\sqrt{3})$ where p and q are integers.

[8 marks]

[illegible]

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Answer

8

10

are α , β , γ and δ , where p and q are real.

It is given that α , β , γ and δ form an arithmetic sequence.

10 (a)

[6 marks]

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Answer _____

[3 marks]

$p =$ $q =$

9



11

$$2x - 12 = \frac{y - 7}{2} = 5 - z$$

The line L_2 has equation

$$\left(\mathbf{r} - \begin{bmatrix} 8 \\ w \\ 6 \end{bmatrix} \right) \times \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix} = \mathbf{0} \quad \text{where } w \text{ is a constant.}$$

11 (a)

Find the value of w

[5 marks]

[illegible]

Answer



[3 marks]

[illegible]

Answer

Question 11 continues on the next page

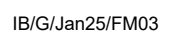
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[4 marks]

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Answer



11 (c) (ii) Find the exact distance of P from Π

[2 marks]

Answer _____

14

Turn over for the next question

Turn over ►



It is given that $y = f(x)$ satisfies the differential equation

and when $x = 0$ it is given that both $y = 10$ and $\frac{dy}{dx} = 1$

[12 marks]

[illegible]

$f(x) =$

12

Turn over ►



13

$$\mathbf{M} = \begin{bmatrix} 4 & 1 & -(k+1) \\ 1 & 3 & k+2 \\ 2 & 1 & 1 \end{bmatrix}$$

where k is a constant.

It is given that $\mathbf{M}$ is a non-singular matrix.

13 (a) (i) Find any restrictions on the value of k

[2 marks]

Answer

13 (a) (ii) Find $\mathbf{M}^{-1}$ in terms of k

[5 marks]



- 13 (b)** Use your answer to **part (a)(ii)** to solve

$$4x + y - (k+1)z = 4$$

$$x + 3y + (k+2)z = 3$$

$$2x + y + z = 1$$

Give your answer in terms of k

[3 marks]

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$x =$ _____ $y =$ _____ $z =$ _____

10

Turn over ►



[6 marks]

[illegible]

Answer _____



[3 marks]

$$\int_0^{\frac{\pi}{6}} \sin^6 \theta \, d\theta = \frac{1}{96} (a\pi + b\sqrt{3})$$

where a and b are integers.

[illegible]

Turn over for the next question

15

$$\frac{dy}{dx} \cos x + y \sin x = \frac{2x+5}{x^2+4x+5} \cos^2 x$$

can be written in the form $y = f(x)$

15 (a)

[9 marks]

[illegible]

$$f(x) =$$

15 (b) Find the particular solution of the differential equation where $f(0) = \tan^{-1}(2)$

[2 marks]

$$y =$$

END OF QUESTIONS



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